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My Contribution to the YOLOv11-Based Animal Image Classification Project

1. Understanding the Research Paper

First, I carefully read the research paper to understand the main goal of the project and how image classification works. The project focuses on classifying five types of animals: dogs, cows, cats, lambs, and zebras. I learned how a YOLOv11-based model can use image features to identify different animal classes.

2. Dataset Collection

One of my main contributions was collecting images for the animal classification dataset. I collected pictures of different animals from various online sources. I also took some pictures myself using my mobile phone. I wanted to include images from different sources to make the dataset more diverse.

3. Dataset Preparation and Classification

After collecting the images, I organized them into separate folders based on their animal classes. I placed dog images in the dog folder, cow images in the cow folder, cat images in the cat folder, lamb images in the lamb folder, and zebra images in the zebra folder.

I checked the images and arranged them so that each image belonged to the correct class. This step was important because the model needed correctly labeled images to learn the differences between the animals.

4. Dataset Size and Image Dimensions

According to the research paper, the custom dataset contains 500 images, with 100 images for each of the five animal classes. The images were collected from online sources and mobile phone captures.

The paper also explains that the images were resized to a fixed resolution before model training. However, the exact dimensions and total storage size of my final YOLO dataset need to be checked from the actual dataset files.

5. Preparing the Dataset for YOLOv11

I prepared the classified images for use with the YOLOv11 classification model. Organizing the images by class helped the model understand which category each image belonged to during training.

I collected pictures of different sizes and then converted them into 256 × 256 size.

The prepared dataset was intended to help the model learn the visual differences between the five animal classes and predict the correct class for a new image.

6. Challenges and Learning

During this work, I learned that collecting images is not the only important part of building an image classification project. The images also need to be organized properly and assigned to the correct classes. I also understood that images collected from different sources can have different backgrounds, lighting conditions, sizes, and angles. These differences can affect how the model learns and makes predictions.

7. Conclusion

My main contribution to this project was collecting animal images from online sources and my own mobile phone, organizing the images into five classes, and preparing the dataset for YOLOv11-based image classification. Through this work, I gained practical experience in dataset collection and preparation. It also helped me understand the importance of a well-organized dataset in a deep learning project.